

"Engineering, Procurement, and Construction Management

Target Release	
Epic	M05 – Topography Survey & Soil Testing
Document Status	Draft
Document Owner	Ashish Singh
Designer	
Tech Lead	Jasim
Document Approver	Jasim
QA	

Module	Project Management
ID	M05_EPC_Project_UC_01
Use Case Name	Topography Survey & Soil Testing
Actor(s)	Project Coordinator: Authorized user responsible for creating survey requests, assigning survey agencies, coordinating survey activities, monitoring execution, reviewing submissions, and completing survey activities. Survey Agency: External agency responsible for conducting topography surveys and soil investigations and submitting reports. Project Head: Responsible for survey oversight, progress monitoring, issue resolution, and ensuring completion within defined timelines. System: Handles validations, workflow transitions, agency management, notifications, document management, SLA monitoring, escalation triggers, report management, and audit logging.
Description	This use case enables the Project Team to initiate, manage, monitor, and complete Topography Survey and Soil Testing activities after successful completion of Project Kickoff & Alignment (M04). The process includes survey request creation, site access verification, coordinate capture, survey guideline communication, survey agency assignment, survey execution, report submission, validation of engineering inputs, and storage of survey outputs. The system captures critical survey information including Survey Date, Agency Name, Soil Load-Bearing Capacity, and Contour Data required for engineering design activities. Upon successful completion of survey activities, the project automatically transitions to M06 – Engineering Design & Layout.
Preconditions	• Kickoff Status = Completed. • Project Status Flag = Ready for Survey. • Project record exists in ERP Project Master. • Project Head has been assigned. • Site access has been confirmed. • Site coordinates are available or can be captured. • Survey guidelines and survey requirements are available. • User has Topography Survey & Soil Testing module access rights. • Project is available in the Topography Survey & Soil Testing Dashboard. • Survey Agency exists in Agency Master or can be added using Add Agency functionality. • Supporting project documents required for survey execution are available.
Post conditions	Success Scenario • Survey Request is successfully created with a unique Survey Request ID, and a Survey Agency is assigned. • Site Access details, Coordinates, Survey Guidelines, and Survey Requirements are successfully recorded against the project. • Topography Survey Report and Soil Investigation Report are successfully uploaded, validated, and stored in the project repository. • Survey Date, Agency Name, Contour Data, and Soil Load-Bearing Capacity are successfully captured and maintained as part of the project survey records. • All survey-related documents and supporting attachments are linked to the project and become available for future reference. • Survey Status becomes Completed, and report submission history is maintained.

	• Audit logs are generated successfully for all survey-related activities. • Project automatically transitions to M06 – Engineering Design & Layout. Failure Scenario • Survey Request creation fails due to validation errors or incomplete mandatory information. • Site Access details, Coordinates, Survey Guidelines, Survey Requirements, or Agency Assignment information are missing. • Mandatory survey data required for execution is incomplete, invalid, or not captured. • Topography Survey Report and/or Soil Investigation Report are not submitted or fail validation. • Uploaded files fail validation due to unsupported file formats, file size restrictions, or incomplete document uploads. • Survey Date, Agency Name, Contour Data, or Soil Load-Bearing Capacity are not captured as part of the project survey records. • Survey activities remain in Draft, In Progress, Pending Submission, or Pending Review status. • Project cannot transition to M06 – Engineering Design & Layout until all mandatory reports, survey data, and validations are successfully completed.								
Assumptions	1. Project Kickoff & Alignment (M04) has been completed successfully and the project is formally approved for survey activities. 2. Site access permissions required for conducting Topography Survey and Soil Investigation activities have been obtained from the client before survey initiation. 3. Project information maintained in the Project Master are accurate and available for survey planning and execution. 4. Survey requirements and survey guidelines are defined by the Project Team and communicated to the assigned Survey Agency before survey execution. 5. Survey Agencies may be assigned to perform Topography Survey, Soil Investigation, or both activities depending on project requirements and scope of work. 6. Survey reports, supporting documents, and related project records submitted during the survey process are treated as official project documents and retained for future engineering, audit, and compliance purposes. 7. Approved survey outputs, including Survey Date, Agency Name, Contour Data, and Soil Load-Bearing Capacity, are mandatory inputs for M06 – Engineering Design & Layout. 8. System automatically maintains audit logs for all Project Survey activities.								
Main Flow	Step 1: User logs into the system using valid credentials and navigates to Projects → Topography & Soil Testing Dashboard and system displays Topography & Soil Testing Dashboard along with the standard application header and dashboard components as defined in the Common UI Components & Standards section." The dashboard contains the following module-specific components: Summary Cards <table border="1"> <thead> <tr> <th>Card Name</th><th>Description</th></tr> </thead> <tbody> <tr> <td>Total Requests</td><td>Total Survey Requests created in the system</td></tr> <tr> <td>Pending Assignment</td><td>Requests awaiting Survey Agency assignment</td></tr> <tr> <td>Survey In Progress</td><td>Survey activities currently ongoing</td></tr> </tbody> </table>	Card Name	Description	Total Requests	Total Survey Requests created in the system	Pending Assignment	Requests awaiting Survey Agency assignment	Survey In Progress	Survey activities currently ongoing
Card Name	Description								
Total Requests	Total Survey Requests created in the system								
Pending Assignment	Requests awaiting Survey Agency assignment								
Survey In Progress	Survey activities currently ongoing								

Survey Completed	Completed survey activities
Soil Test In Progress	Soil Investigation activities currently ongoing
Soil Test Completed	Soil Investigation reports submitted
Topography Test In Progress	Topography Survey activities currently ongoing
Topography Test Completed	Topography Survey activities completed
Reports Pending Review	Submitted reports awaiting review and approval

Search Bar: Allows searching using Project ID/Project Name/Agreement Number/Site Location/ Survey Agency

Filters: The system enables users to filter records by following Survey Status/ Topography Status/ Soil Test Status/ Report Status/ Assigned Agency/ Date range
The user can click:

- **Apply** to apply selected filters.
- **Reset** to clear all selected filters and restore the default dashboard view.

The page displays a Survey Request table containing the following columns:

Field	Description
Request ID	Unique Survey Request Number
Project ID	Unique Project Identifier
Project Name	Project Name
Survey Type	Topography / Soil Testing / Topography & Soil
Requested By	User who created the Survey Request
Request Date	Survey Request Creation Date
Assigned Agency	Assigned Survey Agency
Survey Status	Pending Assignment / Draft / In Progress / Completed / Due Soon / Overdue
Topography Status	Current Topography Survey Status

Soil Test Status	Current Soil Investigation Status
Report Status	Pending / Pending Review / Verified / Completed / Not Available
Due Date	Expected Survey Completion Date
Last Updated	Last Modified Date & Time
Notes	Survey Discussion Count
Action	Action Menu (:)

Step 2: User searches or filters survey records from the Dashboard.

Upon clicking Apply, the system displays all records matching the selected search and filter criteria.

Upon clicking Reset, the system clears all applied filters and restores the default dashboard list.

If no records are available, the system displays: "**No survey records available.**"

The survey list refreshes dynamically based on selected search and filter criteria.

Dashboard also updates all summary card counts based on the applied filter conditions.

Step 3: User identifies an eligible project from the Dashboard and selects an action from the Action Menu (:) and system displays the available actions based on the current Survey Status and user permissions.

Survey Status	Available Actions
Pending Assignment	Start Survey, View Survey
In Progress	Continue Survey, Submit Reports, View Survey
Completed	View Survey, View Reports

Action	Description
Start Survey	Initiates Survey Request & Assignment process for an eligible project received from M04
Continue Survey	Opens an existing survey request for modification and progress updates
Submit Reports	Opens Survey & Soil Report Submission screen
View Survey	Opens Survey Request details in read-only mode
View Reports	Opens uploaded Topography Survey Report and Soil Investigation Report in read-only mode

Step 4: User selects Start Survey from Action Menu (:) then system validates the selected project and opens the Survey Request & Assignment screen.

The system performs the following validations:

- Project exists in Project Master.
- M04 – Project Kickoff & Alignment is completed.
- Project Status = Ready for Survey.
- User has permission to initiate survey activities.
- Survey Request is not already completed for the selected project.

If validation fails, the system displays the applicable validation message and prevents survey initiation.

If validation succeeds, the system opens the Survey Request & Assignment screen which contains the following sections.

Step 4.1: System displays Request Information section containing survey request details and system automatically generates and populates survey request information for the selected project. This section contains the following fields:

Field	Description	Field Type	Source / Behavior
Survey Request ID	Unique Survey Request Number	Text	Auto-generated by the system
Request Date	Survey Request Creation Date	Date	Auto-populated with current system date
Requested By	User creating the survey request	Text	Auto-populated from logged-in user profile
Status	Current Survey Request Status	Text	Auto-populated after successful submission

Step 4.2: The system shall display the associated project information from the Project Master in read-only mode, including Project ID, Client Name, Agreement No., Site Location, Capacity (MW), Project Head and Overall Status. The displayed information shall be automatically populated and shall not be editable by the user.

Step 4.3: System displays the Site Information section with below fields by which user enters survey site details required for survey planning and execution.

Field	Description	Field Type	Mandatory
Site Address	Survey Site Address	Text Area	Yes
GPS Coordinates (Latitude)	Site Latitude	Decimal	Yes

	GPS Coordinates (Longitude)	Site Longitude	Decimal	Yes
	Land Area (Acre)	Total Site Area	Numeric	Yes
	Site Access Availability	Site Access Status	Dropdown	Yes
	Error Messages			
	"Site Address is required." "GPS Coordinates (Latitude) is required." "GPS Coordinates (Longitude) is required." "Land Area is required." "Site Access Availability is required."			
	Step 4.4: System displays the Survey Requirements section with below fields by which user enters survey execution requirements and instructions for the assigned Survey Agency.			
	Field	Description	Field Type	Mandatory
	Survey Type	Topography / Soil Testing / Topography & Soil	Dropdown	Yes
	Survey Priority	High / Medium / Low	Dropdown	Yes
	Expected Completion Date	Planned Survey Completion Date	Date Picker	Yes
	Survey Guidelines / Instructions	Survey execution instructions	Text Area	Yes
	Error Messages			
	"Survey Type is required." "Survey Priority is required." "Expected Completion Date is required." "Survey Guidelines / Instructions are required."			
	Step 4.5: System displays the Survey Agency Assignment section with below fields by which user selects an existing Survey Agency from the Survey Agency dropdown. Upon selecting a Survey Agency, system retrieves and displays agency contact information.			
	Field	Description	Field Type	Mandatory
	Survey Agency	Assigned Survey Agency	Dropdown	Yes
	Contact Person	Agency Contact Person	Text	Yes
	Contact Number	Agency Contact Number	Numeric	Yes
	Email Address	Agency Email Address	Email	Yes
	- Contact Person, Contact Number, and Email Address are automatically populated based on the selected Survey Agency. - Users may update the displayed contact information if required.			
	Validation: Contact Number must contain valid numeric values and Email Address must be in a valid email format.			

Error Messages

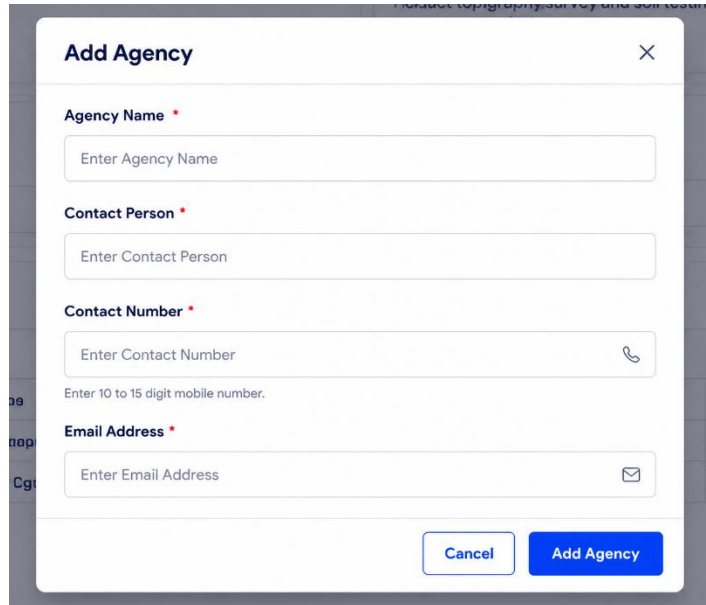
- "Survey Agency is required."
- "Contact Person is required."
- "Contact Number is required."
- "Email Address is required."
- "Please enter a valid Contact Number."
- "Please enter a valid Email Address."

Step 4.6: If the required Survey Agency is not available in the Survey Agency dropdown, the user clicks Add Agency and system opens the Add Agency modal which contains the following fields:

Field	Description	Field Type	Mandatory
Agency Name	Survey Agency Name	Text	Yes
Contact Person	Agency Contact Person	Text	Yes
Contact Number	Agency Contact Number	Numeric	Yes
Email Address	Agency Email Address	Email	Yes

Error Messages

- "Agency Name is required."
- "Contact Person is required."
- "Contact Number is required."
- "Contact Number must contain 10 to 15 digits."
- "Email Address is required."
- "Please enter a valid Email Address."



Upon clicking **Add Agency**:

1. System validates the entered information.

2. A new Agency record is created in Agency Master.
3. The newly created Agency becomes available in the Survey Agency dropdown.
4. The system automatically selects the newly created Agency.
5. Agency contact details are auto-populated in the Survey Agency Assignment section.
6. The activity is recorded in the audit log.

Upon clicking Cancel, the system closes the Add Agency modal without saving any information.

Step 4.7: System displays the Document Uploads section containing the uploaded document list and user clicks Upload Document and system opens the Standard Upload Document Modal as defined in the Common UI Components & Standards section by which user uploads supporting documents required for survey execution, such as Site Layout Plan, Client Site Access Letter, Survey Instructions, Site Drawings, or any other applicable survey reference document.

The uploaded document list displays the following columns:

Field	Description	Field Type	Source / Behavior
#	Serial number	Numeric	System-generated
Document Name	Uploaded document name	Text	Auto-populated from uploaded file
Document Type	Selected document category	Text	User selection from Document Type Master
Uploaded By	User who uploaded the document	Text	System-generated
Uploaded On	Document upload date and time	Date & Time	
File Size	Uploaded file size	Numeric	
Action	View / Download / Delete	Action Menu (:)	System-controlled based on user permissions

Supported File Types= PDF/DOC/DOCX/JPG/PNG

Maximum File Size= 10 MB per file

Validation Rules

- Document Type is mandatory in the upload modal.
- File upload is mandatory before saving the document.
- File format must be PDF, DOC, DOCX, JPG, or PNG.
- File size must not exceed 10 MB.
- User may upload multiple supporting documents.

Error Messages

- "Document Type is required."
- "Please select a file to upload."
- "Invalid file format."

	• "File size exceeds the maximum allowed limit." • "Document upload failed. Please try again." Step 4.8: System displays the Internal Notes section with Remark field by which user enters project-specific remarks, survey observations, special instructions, or internal comments relevant to survey execution. Validation: Remarks cannot exceed the configured character limit. Error Messages: "Remarks cannot exceed 500 characters." Step 4.9: User clicks Save as Draft. The user may choose to save the survey request without assigning it immediately. Before saving the draft, the system performs the following validations: • Project Information must exist. • At least one survey detail must be entered OR • At least one supporting document must be uploaded. If validation passes, the system performs the following actions: • Saves the survey request as Draft. • Updates Survey Status = Draft. • Keeps the project in M05 stage. • Does not assign the Survey Agency. • Creates an audit log entry. Displays confirmation message: "Survey Request draft saved successfully." Step 4.10: After completing all mandatory sections, the user clicks Assign & Submit and system performs the following validations: Validation: • Site Information is completed. • Survey Requirements are completed. • Survey Agency Assignment is completed. • Mandatory supporting documents are uploaded, if configured. • All mandatory fields are completed. If validation fails, the system displays the corresponding validation messages and prevents submission. If validation passes, the system performs the following actions: • Creates the Survey Request. • Confirms the Survey Request and updates its status. • Assigns the selected Survey Agency. • Updates Survey Status = In Progress. • Updates Assigned Agency information. • Records Request Date and Assignment details. • Generates notifications to the assigned Survey Agency and Project Head. • Creates an audit log entry.
--	---

Displays confirmation message: **"Survey Request assigned successfully."**

Step 4.11: User selects **Continue Survey** from the Action Menu (:) and system opens the Survey Request & Assignment screen and loads previously saved information.

User can:

- Modify survey details.
- Update Survey Agency information.
- Upload additional supporting documents.
- Update Internal Notes.
- Save as Draft.
- Click Assign & Submit after completing all mandatory information.

Step 4.12: Upon successful survey assignment, the system performs the following actions:

- Updates survey records in the database.
- Updates dashboard summary card counts.
- Updates Assigned Agency information.
- Updates Due Date tracking.
- Initiates SLA monitoring for Topography Survey and Soil Testing activities.
- Maintains assignment history.
- Creates audit log entries containing: User ID, Survey Request ID, Assigned Agency, Action Performed, Remarks (if any) and Timestamp

After Step 4.12, flow moves to the **A4 – Survey & Soil Report Submission**

Good Morning, Ashish Singh

Survey & Soil Report Submission

Monday, 16 June 2026 | 10:45 AM

Projects > Topography & Soil Testing > Survey & Soil Report Submission

Upload completed survey and soil investigation reports. You can submit any report independently. Marked fields are mandatory

Select Report Type to Submit

Report Type *

Select Report Type

Topography Survey Report

Soil Investigation Report

Both Reports

Project Information (Read Only)

Survey Request ID	Project ID	Client Name	Agreement No.	Site Location
SUR-2026-00125	PRJ-1056	ABC Industries Ltd.	AGR-1056	Jaipur, Rajasthan
Survey Agency	Survey Date	Soil Test Date	Overall Status	
Geo Survey India	06-Jun-2026	08-Jun-2026	In Progress	

Topography Survey Report

Topography Survey Request ID (Auto generated after submit)

Survey Completion Date *

Contour Interval (m) *

Total Survey Area (Acres) *

16-Jun-2026 10:31 AM

0.5

12.45

Uploaded Documents

#	File Name	Uploaded On	Uploaded By	File Size	Action
1	Topography Survey Report.pdf	16-Jun-2026 10:31 AM	Ashish Singh	1.25 MB	View, Download, Delete

Soil Investigation Report

Soil Investigation Request ID (Auto generated after submit)

Soil Test Completion Date *

Number of Boreholes *

Borehole Depth (m)

Soil Bearing Capacity (kN/m²)

Water Table Depth (m)

16-Jun-2026 10:32 AM

3

15.0

250

6.5

Uploaded Documents

#	File Name	Uploaded On	Uploaded By	File Size	Action
1	Soil Investigation Report.pdf	16-Jun-2026 10:32 AM	Ashish Singh	1.48 MB	View, Download, Delete

[Back](#) [Submit Report](#)

Step 5: User identifies a survey request with Survey Status = In Progress and selects Submit Reports from the Action Menu (:) and system opens the Survey & Soil Report Submission screen and retrieves the survey request details associated with the selected project.

The screen contains the following sections:

Step 5.1: user selects the report type from the Report Type dropdown and system displays the Select Report Type to Submit section with the following available options.

- Topography Survey Report
- Soil Investigation Report
- Both Reports

Validation: Report Type is mandatory.

Error Messages: "Report Type is required."

Step 5.2: System displays associated project information from the Project Master in read-only mode, including Survey Request ID, Project ID, Client Name, Agreement No., Site Location, Survey Agency, Survey Date, Soil Test Date, and Overall Status. The displayed information shall be automatically populated and shall not be editable by user.

Step 5.3: System displays the Topography Survey Report section by which user enters Topography Survey details and uploads the completed report , user clicks Click to Upload and system open the Standard Upload Document Modal as defined in the Common UI Components & Standards section.

After successful upload, the system displays the uploaded document in the Uploaded Documents list.

The section contains the following fields:

Field	Description	Field Type	Mandatory
Topography Survey Request ID	Unique Topography Survey Reference	Text	Auto-generated after submission
Survey Completion Date	Topography Survey Completion Date	Date Picker	Yes
Contour Interval (m)	Contour Interval	Decimal	Yes
Total Survey Area (Acres)	Total Survey Area	Decimal	Yes

Step 5.4: System displays the Soil Investigation Report section by which user enters Soil Investigation details and uploads the completed report , user clicks Click to Upload and system opens the Standard Upload Document Modal as defined in the Common UI Components & Standards section.

After successful upload, the system displays the uploaded document in the Uploaded Documents list.

The section contains the following fields:

Field	Description	Field Type	Mandatory
Soil Investigation Request ID	Unique Soil Investigation Reference	Text	Auto-generated after submission
Soil Test Completion Date	Soil Investigation Completion Date	Date Picker	Yes
Number of Boreholes	Total Boreholes Conducted	Numeric	Yes
Borehole Depth (m)	Borehole Depth	Decimal	Yes
Soil Bearing Capacity (kN/m ²)	Soil Bearing Capacity	Decimal	Yes
Water Table Depth (m)	Water Table Depth	Decimal	Yes

After successful upload, the system displays the uploaded document in the Uploaded Documents list.

The uploaded document list contains the following information:

Field	Description
#	Serial Number
File Name	Uploaded Report Name
Uploaded On	Upload Date & Time
Uploaded By	User who uploaded the report
File Size	Uploaded File Size
Action	View / Download / Delete

Step 5.5: After completing the required report details and uploading the applicable report(s), the user clicks Submit Report and system validates the following submitted information.

- Report Type is selected.
- Mandatory report details are completed based on the selected Report Type.
- Required report documents are uploaded.
- Survey Completion Date and/or Soil Test Completion Date are provided.
- All mandatory fields are completed.

If validation fails, system displays the corresponding validation message and prevents report submission.

If validation succeeds, the system submits the selected report(s) successfully and display Success Message: "Survey report submitted successfully."

The system performs the following actions:

	• Updates Survey, Topography, Soil Test and Report Status. • Stores submitted reports in the project repository. • Creates audit logs. • Makes survey outputs available for M06. • Survey Request becomes available in View Survey mode. • Submitted reports become available through View Reports. • Project becomes available in M06 – Engineering Design & Layout work queue.
Alternative Flows	AF-01: If no survey requests are available on the Topography & Soil Testing Dashboard, the system displays the message "No survey records available." AF-02: If the survey activities exceed the configured Expected Completion Date, the system automatically updates the Survey Status to Due Soon or Overdue based on the configured SLA rules, updates the dashboard, and generates notifications/escalations as configured. AF-03: If the user submits only one report (Topography Survey Report or Soil Investigation Report), the system updates the corresponding report status while keeping the Survey Request In Progress until all mandatory reports required for the selected Survey Type are submitted. AF-04: If a system or network error occurs during survey request creation, draft save, agency creation, document upload, survey assignment, or report submission, the system displays an appropriate error message and the transaction is not completed until the user retries the operation.